

Georges Parfait Djimefo

Kapen | Computer Scientist | AI Researcher

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Profile

Computer scientist with expertise in **AI, Machine Learning, Multi-Agent Reinforcement Learning, cybersecurity**, and **6G network systems**. PhD expected Summer 2026. Strong background in software engineering, data science, cloud computing, and applied statistics. Proven ability to design novel AI architectures, author peer-reviewed research, teach at the university level, and deliver production-grade ML systems on AWS. Innovation, teamwork, leadership, and results-oriented.

Technical Skills

Languages: Python, Java, C++, PHP, SQL, R, JavaScript, TypeScript

LLM / Agents: LangChain, LangGraph, LangSmith, MCP, RAG (FAISS), CRAG, RAGAS, Claude (Haiku, Opus), Kiro

AI / ML: PyTorch, TensorFlow, Scikit-Learn, MARL (QMIX, QTRAN), GNN, DQN, NLP, Computer Vision, LLMs

Data: PySpark, Databricks, Snowflake, Delta Lake, Relational Databases, ETL/ELT

Cloud: AWS (SageMaker, Bedrock, EKS, Lambda, CloudWatch, S3, ECR, CloudFormation), OpenStack, Kubernetes, Docker

DevOps: CI/CD (GitHub Actions, CodeBuild, OIDC), Git, Jenkins, GitLab, Terraform

Web / Mobile: React, Angular, Node.js, FastAPI, Streamlit, D3.js, Plotly, Dash, Power BI, Tableau

Methods: Agile/Scrum, TDD, Design Patterns, Statistical Modeling, MLOps, Infrastructure as Code

Publications

[1] *Energy-Efficient and Near Real-Time Routing Algorithm for Disaster Monitoring in Wireless Sensor Networks*

G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal

(Submitted)

- Designed **GAPF**, a novel **meta-learning** framework fusing MARL policies (QMIX & QTRAN) via a **GCN encoder** with only **1 473 parameters**.
- Achieved $\geq 98\%$ packet delivery, $2\text{--}3.5\times$ less energy, $13\times$ less memory than baselines.

[2] *Carbon-Aware Autonomous Data Reduction and Self-Healing Routing for 6G-Integrated Disaster Sensor Networks*

G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal

(Submitted)

- Proposed **CALASH**, the **first protocol** jointly optimizing Lifecycle Carbon Intensity via a hybrid **Lyapunov-DQN** engine, **MAPE-K** self-healing, and ISO 14040 lifecycle analysis over 6G.
- Achieved 60% longer network lifetime and 33% lower LCI than baselines.

[3] *Zero-Trust Shielded Graph-Structured Decision Control for Secure Autonomous Recovery in AI-Native 6G Networks*

G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal

(Submitted)

- Designed **CASTER-ZT**, the **first framework** combining **GNN-based** decision making with a formally bounded **zero-trust** admissibility shield (10 proven properties).
- Achieved 74–98% rogue detection with **zero false positives**, 3 ms latency within O-RAN 10 ms budget.

Professional Experience

Polytechnique Montréal

Lecturer

Montréal

2025–2026

- Teach the Python procedural programming course.

Intact Financial Corporation

AI Developer I

Montréal

2022–2024

- Ingest and process large-scale data.
- Monitor pipelines and develop automated testing solutions.
- *Stack:* PySpark, SQL, Databricks, Snowflake, AWS, Shell, Git.

Polytechnique Montréal

Lecturer

Montréal

2022–2023

- Teach the Python procedural programming course.

Université de Montréal

Scientific Assistant — MOOC AI

Montréal

2021–2022

- Design courses and practical work in supervised learning (Python).

Ministry of Tourism and Recreation <i>Contractual Statistical Economist Engineer</i> ○ Analyze data	Cameroon 2012–2014
Protestant University of Central Africa <i>Associate Professor of Statistics & Probability</i> ○ Teach statistics and probability courses.	Cameroon 2010–2012
ISSEA (Sub-regional Institute of Statistics and Applied Economics) <i>Part-time Lecturer in Data Analysis</i> ○ Teach the Data Analysis course (R).	Cameroon 2009–2010

Internships & Selected Projects

Ericsson <i>Machine Learning Intern</i> ○ Designed and deployed a production RAG Autonomous Agentic AI system for telecom radio unit log anomaly detection, serving Ericsson's enterprise clients. ○ Trained a custom Dual-Attention SLM achieving F1 90.6% , 99.97% accuracy, and 1 ms/log inference latency on AWS SageMaker. ○ Built a 6-tool ReAct agent orchestrated by Claude Haiku 4.5 on Bedrock: SageMaker scoring, FAISS RAG, RF physics world model, pattern classification, CloudWatch health monitoring. ○ Implemented self-correcting RAG (CRAG) with quality grading, RAGAS faithfulness/relevancy evaluation, and evidence-based meta-cognition confidence scoring (HIGH/MEDIUM/LOW). ○ Engineered a production-scale map-reduce pipeline : triage → cluster by failure signature → parallel agent investigation → adaptive cross-cluster synthesis. Handles 120+ anomalies in ~12 min. ○ Deployed the agent as a K8s pod on EKS exposed via Cloudflare tunnel with a Streamlit UI accessible from any device on the corporate network (phone, laptop, browser). ○ Built an industry-standard event pipeline : S3 upload → Lambda → SQS → K8s worker (full agent) → results in S3 → SNS email alerts on critical anomalies. ○ Designed a fully autonomous retraining loop : drift-triggered + weekly retraining via EventBridge, with auto-promotion only when the new model's F1 exceeds the current production model. ○ Implemented a 7-stage CI/CD pipeline (GitHub Actions + AWS CodeBuild with OIDC): test → OWASP security scan → CloudFormation infra → staging gate → deploy → smoke test → auto-rollback. Zero static IAM keys. ○ Authored 131 automated tests with an AST-based adaptive test guardian ensuring 100% production function coverage. ○ <i>Stack</i> : Python, AWS (SageMaker, Bedrock, EKS, Lambda, SQS, SNS, S3, ECR, CloudFormation, CloudWatch), LangChain, LangGraph, LangSmith, FAISS, MCP, Claude, Kiro, Docker, Kubernetes, CI/CD, Terraform.	Montréal 2026
Intact Financial Corporation <i>Data Engineering Intern</i> ○ Manipulate data (PySpark, SQL, Databricks, Snowflake, AWS).	Montréal 2022
Polytechnique Montréal <i>Undergraduate Research Initiation Intern</i> ○ "Modeling of Voice Recognition in Android systems to secure bank payments." ○ Model with Deep Learning (Python); implement and deploy on Android mobile (Java).	Montréal 2020–2021
Polytechnique Montréal <i>Entrepreneur / Research Initiation Intern</i> ○ "Face Recognition modeling integrating racial and gender bias." ○ Model with Deep Learning (Python).	Montréal 2020
Polytechnique Montréal <i>Integrator Project — IATA</i> ○ "Flight Data eXchange — Develop an initial and contextual flight data assessment tool." ○ Contribute to software implementation (Dash, Python, React, Electron, Axios).	Montréal 2021
Polytechnique Montréal <i>NLP Project</i> ○ "Extraction of keywords from texts as well as their type" (Python).	Montréal 2021
Polytechnique Montréal <i>Data Visualization Project — Chaire de recherche en fiscalité et en finances publiques</i> ○ "Develop a web application to compare taxes and provincial and federal tax rates by year." ○ Implement a mockup with Plotly.js, D3.	Montréal 2021
FLEX GROUP <i>Programming Intern</i> ○ "Modeling of Automatic Optical Character Recognition." ○ Implement models in Python, PHP.	Laval 2019–2020
International Bank of Cameroon for Savings and Credit <i>Intern for the Master's in Applied Statistics</i> ○ "Initiation of mathematical models in forecasting a bank's income and expenses" (R).	Cameroon 2006

Education

Polytechnique Montréal

PhD in Computer Engineering

- AI-driven routing, carbon-aware networking, and zero-trust security for 6G sensor networks.

Montréal, Canada

Summer 2026

Polytechnique Montréal

Engineering Diploma in Software Engineering

- Concentration: Artificial Intelligence — Data Sciences.

Montréal, Canada

Advanced National School of Engineering (ENSPY)

Master's Degree in Applied Statistics

- Equivalency: Master's in Probability — Statistics.

Yaoundé, Cameroon

University of Yaoundé I

Master's Degree in Mathematics

- Equivalency: Specialized Higher Studies Diploma (DESS) in Mathematics.

Yaoundé, Cameroon

Licenses & Certifications

- AWS Certified Cloud Practitioner (CLF-C01)
Certification Prep Path (Pluralsight), Montréal
- Program Management Professional (PgMP)
Certification Prep Path (Pluralsight), Montréal
- AWS Certified Developer — Associate (DVA-C02)
Certification Prep Path (Pluralsight), Montréal

Conferences & Seminars

PolyCongrès 2026, Polytechnique Montréal

Oral Presentation

“Energy-Efficient and Near Real-Time Routing for Disaster Monitoring in Wireless Sensor Networks — Graph-Attentive Policy Fusion (GAPF).” Georges Parfait Djimefo, Ranwa Al Mallah, Samuel Pierre.

Montréal, Canada

Mar–Apr 2026

20th Conf. African Assoc. of Insect Scientists

Oral Presentation

Djimefo P., Tchuente M. & Le Gall P. — “Biogeography of communities of insects using clustering methods.” Theme: “Global change and impacts on insect biodiversity: Integrated pest and disease vectors management in Africa.”

Yaoundé, Cameroon

Oct 2013

29th Annual Conf. Int. Society for Clinical Biostatistics

Poster

“HIV incidence estimation from repeated cross-sectional surveys: a comparative study of parametric and non-parametric approaches by simulation.” Joint work: H. Gwet, R. Pouillot, Y. Le Strat, A. Alioum, J.-C. Thalabard.

Copenhagen, Denmark

Aug 2008

ANMSA (African Network of Mathematical Statistics and its Applications)

Oral Presentation

Université des Sciences et Techniques de Masuku (USTM). “Épidémiologie du VIH au Cameroun: Estimation de l'incidence à partir des données de prévalence.”

Franceville, Gabon

Jan 2008

Volunteering

- Volunteer at the IEEE branch of Polytechnique Montréal, Montréal, 2018.
- Volunteer at the ConFoo conference, Montréal, 2018.